ASPECT WAVE LLC

Vessel Measurement Recorder

November 2026 · Prepared by Sophie Scopazzi · Firmware v8.12

Equations Used by Firmware

The following are all the equations used by the firmware in calculating values. Angles are in degrees unless marked; lengths in meters.

Sign Conventions

With the BNO085 flat, chip side up and its X arrow toward the bow (Y to port, Z up):

Quantity	Positive
φ roll	Starboard side down (shown S); negative = port down (P)
θ pitch	Bow up
Yaw	Clockwise from above, like a heading. Relative (gyro and accelerometer, no magnetometer), so it drifts slowly
Rate of turn	Turning to starboard, °/min
acc_fwd, acc_stbd, acc_up	Felt force (gravity included) in ship axes, g. Level and still = 0, 0, +1

If one reads backwards, set ROLL_FLIP, PITCH_FLIP or YAW_FLIP to 1 in the sketch.

Level Zero

Roll and pitch run through a 5-s running mean. Set level zero stores that mean as the offset, so

$$\varphi_{shown} = \varphi_{raw} - \varphi_0 \qquad \theta_{shown} = \theta_{raw} - \theta_0$$

The card 16 arrows nudge the offsets 0.1° per tap. Use it only with the vessel upright and still: any list or trim at that moment is removed from every reading.

Roll Period (Card 1)

The roll period as felt, timed one roll at a time:

- Roll is low-passed by two first-order stages (τ = 0.15 s each) to remove vibration.
- A mean that follows a slowly changing list without lagging is removed (60 s):

$$m_1 \leftarrow m_1 + k(r - m_1) \qquad m_2 \leftarrow m_2 + k(m_1 - m_2) \qquad k = \frac{\Delta t}{60 + \Delta t}$$

$$x = r - (2m_1 - m_2)$$

- A roll is the time between upward zero-crossings of x, interpolated between samples. It counts only if x swung past ±0.3° both ways and lasted 2–60 s.
- The roll period is the median of the last 9 rolls (at least 3). It clears after 60 s without a roll or after a data gap.

Rolls timed one by one include rolling forced at the wave encounter period, so this reads shorter than the natural roll. GM and the Te / Tr check use the roll spectrum peak instead.

Roll Statistics (Card 5)

Over the last 10 or 30 min, each roll counted above gives one amplitude, half its peak-to-peak swing about the running mean:

$$A_i = \frac{x_{max} - x_{min}}{2}$$

The significant roll is the mean of the highest third, like significant wave height:

$$\varphi_{1/3} = \frac{1}{k}\sum_{i=1}^{k} A_{(i)} \qquad k = \lceil n/3 \rceil$$

with the amplitudes sorted largest first. The mean list and RMS roll come from 10-s sums of roll and roll²:

$$\varphi_{mean} = \frac{S_1}{N} \qquad \varphi_{RMS} = \sqrt{\frac{S_2}{N} - \varphi_{mean}{}^2}$$

Draft

One draft is used for GM, sea-level pressure, freeboard, air draft and the hull-form checks: the quarter mean when all three drafts are entered (v8.12), as a sagging hull displaces more than the mean of the ends says:

$$d = \frac{F + 6M + A}{8} \text{ with a midships draft;} \qquad \text{otherwise } d = \frac{F + A}{2}$$

where

Symbol	Meaning
F, M, A	Drafts forward, midships and aft read at the marks
d	The draft used (m)

GM Estimate (Card 5)

The IMO rolling-period formula (IS Code 2008, Part A, 2.3):

$$T = \frac{2CB}{\sqrt{GM}} \quad \Rightarrow \quad GM = \left(\frac{2CB}{T}\right)^2$$

$$C = 0.373 + 0.023\frac{B}{d} - 0.043\frac{L}{100}$$

where

Symbol	Meaning
B	Beam (m), card 15
d	Draft (m), as above
L	LBP (m), or LOA if there is no LBP
T	Natural roll period (s): the peak of the roll spectrum (Waves), not the card 1 roll period
GM	Metacentric height (m); shown in feet (× 3.2808) with metres beside it

No GM without the beam, a draft and a length; for the first 8.5 min of steady motion; at the dock, paused or with the IMU off; or from an old wave estimate. With the roll peak within ±15 % of Te, GM is still given, noted “Te ≈ Tr: near resonance”.

Why the spectrum peak: on 1 November 2026 (loaded) the card 1 roll period was 13.7 s against a 14.6 s spectrum peak, giving 18–19 ft and 15.4 ft against CargoMax’s 14.48 ft. Overnight on 3 November the mean was 14.14 ft. A cross-check only, not a stability calculation.

Wave Analysis (Card 6)

Every 0.5 s the VMR keeps 0.5-s means of the vertical acceleration in the earth frame, pitch and roll. The acceleration is the accelerometer turned by the orientation quaternion, less gravity:

$$a_z = (R(q)\,a)_z - g, \quad \text{g = 9.80665 m/s²}$$

The window holds the last 2048 samples (17.07 min). A gap, a pause, an IMU dropout or Dock mode starts it again. From 1024 samples (8.5 min) on, Welch spectra are worked out every 30 s: segments of N = 256 samples (128 s), half overlapped, each with its mean removed and a Hann window, then an FFT:

$$w_i = 0.5 - 0.5\,cos\frac{2\pi i}{N}$$

$$S(f_k) = \frac{2|X_k|^2}{f_s \sum {w_i}^2} \cdot \frac{1}{n_{seg}} \qquad \Delta f = \frac{f_s}{N} = \frac{1}{128}\,\text{Hz}$$

with fs = 2 Hz. Bins k = 4 to 80 are used: periods of 32 s down to 3.2 s. The peak of each spectrum is refined with a parabola through the largest bin (b) and its neighbours (a, c):

$$\delta = \frac{a - c}{2(a - 2b + c)} \qquad T_p = \frac{1}{(k_b + \delta)\Delta f}$$

Period	From
T_e	Peak of the pitch spectrum: the pitching period, used as the wave encounter period (IMO MSC.1/Circ.1228, 1.6)
T_r	Peak of the roll spectrum: the natural roll period
T_z	Peak of the heave spectrum

Synchronous and Parametric Rolling

SYNC! if $\left|\frac{T_e}{T_r} - 1\right| \le 0.15$ PARAM! if $\left|\frac{T_e}{0.5T_r} - 1\right| \le 0.15$

MSC.1/Circ.1228, 4.2.3 names these conditions but gives no tolerance; ±15 % is the VMR’s choice.

Heave Hs at the VMR

The heave spectrum comes from the acceleration spectrum by dividing by (2πf)⁴. Below 0.05 Hz (periods over 20 s) it is set to zero, as integrating there only amplifies noise:

$$S_z(f) = \frac{S_a(f)}{(2\pi f)^4} \qquad m_0 = \sum_k S_z(f_k)\,\Delta f \qquad H_s = 4\sqrt{m_0}$$

Heave Hs at Midships

With the VMR’s position entered in card 15, the hull is taken as rigid. For small angles the VMR rises and falls with the heave at midships plus what pitch and roll give it:

$$z_{VMR} = z_{mid} + x\theta - y\varphi$$

So, frequency by frequency, from each segment's complex spectra (which keep the phase between the three):

$$Z_{mid}(f) = \frac{A(f)}{-(2\pi f)^2} - x\Theta(f) + y\Phi(f)$$

where

Symbol	Meaning
x	VMR distance from midships (m), + forward
y	VMR distance from the centreline (m), + starboard
A, Θ, Φ	Complex spectra of vertical acceleration, pitch and roll (pitch and roll in radians)

|Zmid|² is averaged over the segments and scaled like the other spectra, then Hs at midships = 4√m0 as above. Pitch and roll are averaged over the same 0.5 s as the acceleration so they line up in time. Card 6 shows Hs at midships when the position is entered, otherwise at the VMR. The height above the keel is stored but not used here.

A long hull hardly heaves in waves much shorter than itself, so Hs is a guide to the sea, not the wave height; long swell comes closest. On 3 November, with x = −95 m, Hs at midships read about 0.1–0.7 m higher than at the VMR.

Ship Data Checks (Card 15)

$$V = L\,B\,d \qquad C_b = \frac{1.01605\,\Delta_{LT}}{1.025\,V} \qquad \Delta_t = 1.025\,C_b\,V$$

$$C_{wp} = \frac{39.03 \cdot TPI}{L\,B}$$

$$\text{hog / sag} = M - \frac{F+A}{2} \qquad \text{freeboard} = D_{deck} - d \qquad \text{air draft} = H_{top} - d$$

where

Symbol	Meaning
V	Box volume at the draft (m³)
Δ_{LT}, Δ_t	Displacement in long tons and in tonnes (1 LT = 1.01605 t; sea water 1.025 t/m³)
TPI	Long tons per inch immersion; 1 LT/in in sea water = 39.03 m² of waterplane
$hog\ /\ sag$	+ = sagging
D_{deck}, H_{top}	Keel to main deck, keel to the very top

Saving points out: a Cb from the displacement outside 0.45–0.92 (is it displacement, not deadweight, and LT / MT right?), or more than 0.06 from the Cb entered; a Cwp from the TPI outside 0.6–1.0; a trim over 3 % of L or a hog / sag over L/250; keel heights out of order.

Trim and List at the Dock (Card 11)

Pitch and roll are averaged into 1-s means, then over the chosen window (30 s to 5 min):

$$t = L\,\tan(\theta - \theta_0) \qquad \text{list} = \varphi_{mean} \qquad \text{across the beam} = B\,\tan|\,\varphi_{mean}\,|$$

Trim t is + by the stern, with L the LBP (or LOA). Calibrating from the drafts read at the marks (F, A in inches) stores the offset that makes the window's mean pitch θmean match them:

$$\theta_{true} = tan^{-1}\frac{0.0254\,(A-F)}{L} \qquad \theta_0 = \theta_{mean} - \theta_{true}$$

Without a calibration the trim follows level zero. 0.1° is L tan 0.1° = 0.49 m, about 1.6 ft on 280 m. The sensor reads the deck slope where it is mounted and hull bending changes that slope near the ends, so recalibrate when the loading changes much.

Weather (Card 4)

Sea-level pressure, by the barometric formula with the standard lapse rate:

$$P_0 = P\left(1 - \frac{0.0065\,h}{T + 0.0065\,h + 273.15}\right)^{-5.257} \qquad h = H_{bridge} - d$$

where

Symbol	Meaning
P	Pressure at the BMP388 (hPa); the CSV keeps this, not the sea-level value
h	Sensor height above the waterline (m): keel to bridge less the draft
T	Air temperature (°C): the SHT30, else the BMP388's own sensor, else 15 °C

Dew point, by the Magnus formula (a = 17.62, b = 243.12 °C), with RH in %:

$$\gamma = ln\frac{RH}{100} + \frac{a\,T}{b+T} \qquad T_d = \frac{b\gamma}{a-\gamma}$$

Where the air temperature meets the dew point, expect fog or condensation. The 3-hour change uses a sample every 10 min:

$$\Delta P_{3h} = P_{now} - P_{-3h} \qquad \text{steady if } |\Delta P_{3h}| < 0.5 \text{ hPa}$$

Graphs keep 2-s means for 20 min and 30-s means for 3 h.

Magnetic Heading Test (Card 1)

The BNO085's calibrated field m (µT) is read at 10 Hz and turned into the earth frame with the orientation quaternion, which removes roll and pitch. The bow (the sensor's X axis) is turned the same way, b. The horizontal field is then split along the bow and to starboard:

$$w = R(q)\,m \qquad fwd = w_x b_x + w_y b_y \qquad stb = w_x b_y - w_y b_x$$

$$\psi_m = atan2(-stb, fwd) \qquad dip = atan2\left(-w_z, \sqrt{{w_x}^2 + {w_y}^2}\right)$$

fwd and stb are smoothed over about 2 s (the components, not the angle, which wraps at 360°). The ship's deviation is not removed, and on a steel hull it can be large: a test only. It does not feed roll, pitch or yaw.

Rate of Turn, G-Meter, SOG and COG

- **Rate of turn:** the gyro's Z rate × 60 (°/min), + to starboard. The screens and maxima smooth it over 1 s and show values inside the deadband as 0; the CSV keeps the raw rate.
- **G-meter:** the felt push in the deck plane (up = forward, right = starboard; gravity included, so a starboard heel puts the dot to starboard), smoothed over 0.3 s, with peak-hold ticks. The bar shows vertical g; 1 g = still.

- **SOG and COG:** from 1-Hz GPS over the averaging time (1–60 s, default 10 s). SOG is the mean speed; COG is the direction of the mean velocity, so it wraps through 000/360 properly and near-zero speeds carry little weight:

$$COG = atan2\left(\sum v \sin c, \sum v \cos c\right)$$

Indicator LED Brightness and Morse

5 kHz PWM, 10 bits. The duty follows the OLED brightness c (1–255, from the knob or the dashboard slider); the square evens out the steps to the eye and 12 keeps the LEDs just visible at the bottom:

$$duty = 12 + 1011\left(\frac{c}{255}\right)^2 \quad \text{(of 1023)}$$

FAULT in Morse is ..-. .- ..- .-.. - at 120 ms a unit (dot 1, dash 3, gaps 1 in a letter, 3 between letters, 7 between words): 52 units, about 6.2 s.

Constants

Constant	Value
Gravity g	9.80665 m/s²
Sea water density	1.025 t/m³
Long ton	1.01605 t
1 LT/in, sea water	39.03 m² of waterplane
Units	1 m = 3.2808 ft; 1 in = 0.0254 m; 1 hPa = 0.02953 inHg
Lapse rate; exponent	0.0065 K/m; 5.257
Magnus a, b	17.62; 243.12 °C
GM coefficient	C = 0.373 + 0.023 B/d − 0.043 L/100
Wave sampling	2 Hz; 2048-sample window (17.07 min); first after 1024 (8.5 min); every 30 s
Welch	256-sample segments (128 s), 50 % overlap, Hann window
Spectrum band	3.2–32 s (bins 4–80); heave cut below 0.05 Hz
Te / Tr tolerance	±15 %
Roll period	±0.3°, 60 s mean, median of 9 rolls; level zero 5-s mean